Monte Carlo Pricing of a European Call: Exact Simulation under Geometric Brownian Motion

Quant Finance Portfolio — Phase 2, Block 1, Algorithm 1

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Abstract

We construct, analyse, and prepare for implementation the standard Monte Carlo estimator of a European call option under geometric Brownian motion, using exact simulation of the terminal asset price. The setting is analytically degenerate: the price admits the Black–Scholes closed form derived in Phase 1, and the algorithm is therefore presented not for its practical necessity but as the canonical Monte Carlo problem against which every more elaborate method of Phase 2 will be measured. Three deliverables emerge. First, the estimator is exhibited and the regularity hypotheses of the Block 0 framework are verified explicitly. Second, the variance of the discounted payoff is computed in closed form, providing both an a priori sample-size criterion and a sharp baseline for the variance reductions of Block 2. Third, a triangulated validation strategy is articulated, anchored on the Phase 1 Black–Scholes formula, on the $O(n^{-1/2})$ asymptotic error rate, and on the agreement between sample variance and its closed-form value.

Contents

1 Introduction **1**

2 Setting and notation **2**
2.1 Risk-neutral dynamics . . . 2
2.2 The European call payoff . . . 2
2.3 The Black–Scholes constants . . . 2

3 Exact simulation of the terminal price **2**
3.1 Closed-form solution of the SDE . . . 2
3.2 The single-normal sampling scheme . . . 3

4 The Monte Carlo estimator **3**
4.1 Definition and basic properties . . . 3
4.2 Verification of the second-moment hypothesis . . . 3
4.3 Confidence interval and stopping criterion . . . 4

5 Closed-form variance of the discounted payoff **4**
5.1 Decomposition of the second moment . . . 4
5.2 The three expectations . . . 4
5.3 The closed-form variance . . . 5
5.4 Implications for sample-size selection . . . 5

6 Validation strategy **5**

7 Algorithmic specification **6**

8 Looking ahead **6**

9 Bibliographic notes **7**

1 Introduction

The European call option under geometric Brownian motion is the most analytically tractable contingent claim in mathematical finance: its price is given by the Black–Scholes formula

$$C_{\mathrm{BS}} \;=\; S_0\, N(d_1) \;-\; Ke^{-rT}\, N(d_2), \tag{1}$$

with the standard definitions of d_1, d_2 recalled in Section 2. Phase 1 of this project derived this formula through the equivalent martingale measure framework and implemented it in Python and C++.

The natural question is therefore why one would price this option by Monte Carlo simulation at all. The honest answer is that, in this setting, one would not—outside of pedagogical or validation contexts. The algorithm developed in this writeup is the simplest possible instance of the Monte Carlo machinery that will support every other method in Phase 2, and its value lies in the role it plays as a foundation for what follows.

That role is threefold. First, the algorithm is the canonical reference implementation of a Monte Carlo *engine*: a function that consumes a model, a payoff, a sample size, and a random source, and returns a point estimate together with a quantified uncertainty. The architecture designed here—the partitioning of responsibilities between sampling, payoff evaluation, and statistical reduction—will recur essentially unchanged throughout the phase. Second, this is the only algorithm of the phase entirely free of systematic error: the exact transition law of the underlying is known and used directly, so no time-discretisation bias enters, and no nested estimation introduces regression bias. Any discrepancy between the estimator and the Black–Scholes price is therefore unambiguously attributable to sampling error, which makes this algorithm an unusually clean diagnostic for implementation correctness. Third, because the distribution of the discounted payoff is known explicitly, its variance admits a closed-form expression, which we derive in Section 5. This expression is the denominator against which the variance reductions of Block 2 will be measured, and it provides an additional independent test of the implementation.

The remainder of the document is organised as follows. Section 2 fixes notation. Section 3 records the closed-form solution of the Black–Scholes SDE and formalises the notion of exact simulation that the title invokes. Section 4 presents the estimator, verifies the second-moment hypothesis required by the Block 0 framework, and specialises the confidence interval to the present setting. Section 5 derives the closed-form variance of the discounted payoff and discusses its implications for sample-size selection. Section 6 articulates the validation strategy. Section 7 states the algorithm formally. Section 8 situates the algorithm within the rest of Phase 2.

2 Setting and notation

2.1 Risk-neutral dynamics

We work on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\in[0,T]}, \mathbb{Q})$ supporting a one-dimensional $\mathbb{Q}$-Brownian motion W. Under $\mathbb{Q}$, the price process of the underlying asset satisfies the stochastic differential equation

$$\mathrm{d}S_t \;=\; r\, S_t\, \mathrm{d}t \;+\; \sigma\, S_t\, \mathrm{d}W_t, \qquad S_0 > 0, \tag{2}$$

where $r \in \mathbb{R}$ is the constant risk-free rate and $\sigma > 0$ is the constant volatility. The numeraire is the money-market account $B_t = e^{rt}$, so that the deflated price S_t/B_t is a $\mathbb{Q}$-martingale, and the discounted payoff of any contingent claim admits a $\mathbb{Q}$-expectation interpretation.

2.2 The European call payoff

A European call with strike $K > 0$ and maturity $T > 0$ pays $(S_T - K)^+$ at time T. Its time-zero price is the discounted expectation

$$C \;=\; \mathbb{E}^{\mathbb{Q}}\big[\, e^{-rT}\,(S_T - K)^+ \,\big] \;=\; \mathbb{E}[\,Y\,], \tag{3}$$

where we have set

$$Y \;\stackrel{\text{def}}{=}\; e^{-rT}\,(S_T - K)^+ \tag{4}$$

and adopt the convention that, where unambiguous, the superscript on $\mathbb{E}$ is dropped.

2.3 The Black–Scholes constants

We use the notation

$$d_1 \;\stackrel{\text{def}}{=}\; \frac{\log(S_0/K) + \big(r + \frac{1}{2}\sigma^2\big)T}{\sigma\sqrt{T}}, \qquad d_2 \;\stackrel{\text{def}}{=}\; d_1 - \sigma\sqrt{T}. \tag{5}$$

The derivation of (1) from (3) via the martingale approach is given in the Phase 1 writeup and will not be repeated.

3 Exact simulation of the terminal price

3.1 Closed-form solution of the SDE

The SDE (2) admits a closed-form solution,

$$S_t \;=\; S_0 \exp\!\big(\big(r - \tfrac{1}{2}\sigma^2\big)t \;+\; \sigma\, W_t\big)\,, \qquad t \in [0, T], \tag{6}$$

which one verifies by applying Itô's lemma to $f(t, x) = \log x$ and integrating. As a consequence, S_T is log-normally distributed: $\log S_T$ is Gaussian with mean $\log S_0 + (r - \frac{1}{2}\sigma^2)T$ and variance $\sigma^2 T$.

3.2 The single-normal sampling scheme

Because $W_T \stackrel{d}{=} \sqrt{T}\, Z$ with $Z \sim N(0, 1)$, exact samples of S_T are obtained by

$$Z \sim N(0, 1), \qquad S_T \;=\; S_0 \exp\!\Big(\big(r - \tfrac{1}{2}\sigma^2\big)T \;+\; \sigma\sqrt{T}\, Z\Big)\,, \tag{7}$$

each replication consuming exactly one standard normal draw. There is no time-discretisation: the construction reaches T in a single step.

Remark 3.1 (The meaning of "exact"). The qualifier "exact" refers strictly to the absence of time-discretisation error in the construction (7): the random variable on the right-hand side has *exactly* the distribution of S_T under the model. It does not refer to the estimator of the price, which carries the usual statistical error inherent to Monte Carlo. The distinction matters because the algorithms of Block 1.2 (Euler–Maruyama, Milstein) will not be exact in this sense: their simulated terminal values will be drawn from a distribution that approximates, but does not coincide with, the true distribution of S_T.

4 The Monte Carlo estimator

4.1 Definition and basic properties

Let $Z_1, Z_2, \ldots, Z_n$ be independent standard normal random variables, and set

$$S_T^{(i)} \;=\; S_0 \exp\Big(\big(r - \tfrac{1}{2}\sigma^2\big)T \;+\; \sigma\sqrt{T}\, Z_i\Big), \qquad Y_i \;=\; e^{-rT}\,\big(S_T^{(i)} - K\big)^+. \tag{8}$$

The Monte Carlo estimator of C based on n paths is

$$\hat{C}_n \;\stackrel{\text{def}}{=}\; \frac{1}{n}\sum_{i=1}^{n} Y_i. \tag{9}$$

By construction, $\hat{C}_n$ is unbiased: $\mathbb{E}[\hat{C}_n] = C$. By the strong law of large numbers, $\hat{C}_n \to C$ almost surely as $n \to \infty$.

4.2 Verification of the second-moment hypothesis

The asymptotic apparatus of Block 0—in particular the central limit theorem and the resulting confidence interval—requires that $\mathrm{Var}(Y) < \infty$, equivalently $\mathbb{E}[Y^2] < \infty$. We verify this explicitly.

Lemma 4.1. *The discounted payoff Y defined in* (4) *satisfies* $\mathbb{E}[Y^2] < \infty$.

Proof. The pointwise inequality $0 \le (S_T - K)^+ \le S_T$ yields $0 \le Y \le e^{-rT} S_T$. Hence $\mathbb{E}[Y^2] \le e^{-2rT}\,\mathbb{E}[S_T^2]$. From (6),

$$\mathbb{E}[S_T^2] \;=\; S_0^2\,\mathbb{E}\Big[\exp\big(2(r - \tfrac{1}{2}\sigma^2)T + 2\sigma\sqrt{T}\,Z\big)\Big] \;=\; S_0^2\,\exp\big((2r + \sigma^2)T\big),$$

where the last equality uses the moment-generating function of the standard normal. Therefore $\mathbb{E}[Y^2] \le S_0^2\, e^{(\sigma^2 - 2r)T} \cdot e^{2rT} = S_0^2\, e^{\sigma^2 T}$, which is finite. □

The hypotheses of the Block 0 framework are therefore satisfied without restriction, and all conclusions of Sections 2.2–2.6 of that writeup specialise directly to the present estimator.

4.3 Confidence interval and stopping criterion

Following the Block 0 prescription, the asymptotic $(1-\beta)$-confidence interval for C is

$$\hat{C}_n \;\pm\; z_{\beta/2}\,\frac{S_n}{\sqrt{n}}, \tag{10}$$

with $S_n^2 = (n-1)^{-1}\sum_{i=1}^{n}(Y_i - \hat{C}_n)^2$ the sample variance of the payoffs and $z_{\beta/2}$ the upper $\beta/2$-quantile of the standard normal. We denote the half-width by h_n. Two natural stopping rules emerge:

1. Fix n in advance (the more common choice in benchmarking work).
2. Continue sampling until h_n falls below a prescribed absolute or relative tolerance.

Both will be supported by the implementation; the validation tests of Section 6 use the first.

5 Closed-form variance of the discounted payoff

We now derive an explicit expression for $\mathrm{Var}(Y)$, which is finite by Lemma 4.1 and computable in closed form because the distribution of S_T is known. The derivation reuses the change-of-measure techniques familiar from the proof of (1) in Phase 1.

5.1 Decomposition of the second moment

On the event $\{S_T > K\}$, the payoff is $S_T - K$; elsewhere it vanishes. Hence

$$Y^2 \;=\; e^{-2rT}\,(S_T - K)^2\,\mathbf{1}_{\{S_T>K\}} \;=\; e^{-2rT}\big[S_T^2\,\mathbf{1}_{\{S_T>K\}} - 2K\,S_T\,\mathbf{1}_{\{S_T>K\}} + K^2\,\mathbf{1}_{\{S_T>K\}}\big]\,. \tag{11}$$

Three expectations must therefore be computed:

$$A \stackrel{\text{def}}{=} \mathbb{E}[\mathbf{1}_{\{S_T>K\}}], \qquad B \stackrel{\text{def}}{=} \mathbb{E}[S_T\mathbf{1}_{\{S_T>K\}}], \qquad D \stackrel{\text{def}}{=} \mathbb{E}[S_T^2\mathbf{1}_{\{S_T>K\}}]. \tag{12}$$

5.2 The three expectations

The first two are obtained as part of the standard derivation of the Black–Scholes formula and we record them without re-deriving:

$$A = N(d_2), \qquad B = S_0\,e^{rT}\,N(d_1). \tag{13}$$

The third expectation is the new quantity. Writing S_T in the form (6) with $W_T = \sqrt{T}Z$,

$$S_T^2 \;=\; S_0^2\exp\big((2r-\sigma^2)T + 2\sigma\sqrt{T}\,Z\big),$$

we have, with $\mu \stackrel{\text{def}}{=} (2r-\sigma^2)T$ and $\nu \stackrel{\text{def}}{=} 2\sigma\sqrt{T}$,

$$D \;=\; S_0^2\,\mathbb{E}\big[e^{\mu+\nu Z}\,\mathbf{1}_{\{Z>-d_2\}}\big]\,, \tag{14}$$

where the equivalence $\{S_T > K\} = \{Z > -d_2\}$ follows from inverting (6) for Z. Now perform the exponential change of measure

$$\frac{\mathrm{d}\tilde{\mathbb{Q}}}{\mathrm{d}\mathbb{Q}} \;=\; e^{\nu Z - \nu^2/2}. \tag{15}$$

Under $\tilde{\mathbb{Q}}$, the random variable $\tilde{Z} = Z - \nu$ is standard normal (this is the elementary form of the Cameron–Martin theorem). Substituting,

$$\mathbb{E}\big[e^{\nu Z}\,\mathbf{1}_{\{Z>-d_2\}}\big] \;=\; e^{\nu^2/2}\,\tilde{\mathbb{E}}\big[\mathbf{1}_{\{Z>-d_2\}}\big] \;=\; e^{\nu^2/2}\,\tilde{\mathbb{P}}(\tilde{Z} > -d_2 - \nu) \;=\; e^{\nu^2/2}\,N(d_2+\nu).$$

Therefore

$$D \;=\; S_0^2\,e^{\mu+\nu^2/2}\,N(d_2+\nu) \;=\; S_0^2\,e^{(2r+\sigma^2)T}\,N\big(d_1+\sigma\sqrt{T}\big), \tag{16}$$

where we used $\mu + \nu^2/2 = (2r-\sigma^2)T + 2\sigma^2 T = (2r+\sigma^2)T$ and $d_2 + \nu = d_2 + 2\sigma\sqrt{T} = d_1 + \sigma\sqrt{T}$.

5.3 The closed-form variance

Substituting (13) and (16) into (11) and combining with $\mathbb{E}[Y] = C_{\mathrm{BS}}$ yields the following result.

Proposition 5.1 (Closed-form variance of the discounted call payoff)**.** *Let Y be defined as in* (4) *under the dynamics* (2)*. Then*

$$\mathrm{Var}(Y) \;=\; S_0^2\,e^{\sigma^2 T}\,N\big(d_1+\sigma\sqrt{T}\big) \;-\; 2KS_0\,e^{-rT}\,N(d_1) \;+\; K^2\,e^{-2rT}\,N(d_2) \;-\; C_{\mathrm{BS}}^2, \tag{17}$$

where d_1, d_2 are as in (5) *and C_{BS} as in* (1)*.*

Proof. Combine (11), (13), and (16) to obtain $\mathbb{E}[Y^2]$:

$$\mathbb{E}[Y^2] \;=\; e^{-2rT}\Big[S_0^2 e^{(2r+\sigma^2)T}N(d_1+\sigma\sqrt{T}) - 2K\cdot S_0 e^{rT}N(d_1) + K^2 N(d_2)\Big]\,.$$

Distributing e^{-2rT} gives the first three terms of (17). The identity $\mathrm{Var}(Y) = \mathbb{E}[Y^2] - \mathbb{E}[Y]^2$ with $\mathbb{E}[Y] = C_{\mathrm{BS}}$ produces the last term. □

5.4 Implications for sample-size selection

Proposition 5.1 permits an a priori calculation of the sample size required to achieve a prescribed half-width ε at confidence level $1-\beta$:

$$n_\varepsilon \approx \left(\frac{z_{\beta/2}}{\varepsilon}\right)^2 \mathrm{Var}(Y). \tag{18}$$

This is one of the few cases in financial Monte Carlo where the required sample size is known before any path is simulated. In every later algorithm of the phase the variance must be estimated from a preliminary run, and (18) becomes a feedback loop rather than a one-shot calculation.

The expression also serves a more lasting purpose: it is the denominator $\mathrm{Var}_{\text{vanilla}}$ of the variance reduction factors

$$\mathrm{VRF} \stackrel{\text{def}}{=} \frac{\mathrm{Var}_{\text{vanilla}}}{\mathrm{Var}_{\text{technique}}}$$

that will be reported throughout Block 2 to quantify the gain of each variance reduction technique.

6 Validation strategy

We follow the triangulation principle established in Phase 1: the algorithm is subjected to several independent tests, each closing a distinct potential failure mode. Passing any one of them is necessary but not sufficient; passing all of them makes incorrectness highly improbable, although strictly speaking unproven.

Test 1: containment of the Black–Scholes price. For a representative grid of parameters (S_0, K, T, r, σ) and a sample size n moderate to large (e.g. $n = 10^4$ to 10^6), generate the estimate $\hat{C}_n$ together with its half-width h_n, and verify that C_{BS} lies within $[\hat{C}_n - h_n, \hat{C}_n + h_n]$. Repeating this experiment over many independent seeds, the empirical containment frequency should approach the nominal confidence level $1-\beta$. This is the simplest end-to-end check and tests the entire pipeline, from sampling to statistical reduction.

Test 2: empirical convergence rate. For a fixed parameter configuration, evaluate $\hat{C}_n$ at sample sizes $n_1 < n_2 < \cdots < n_K$ spanning several orders of magnitude. Plot $\log|\hat{C}_n - C_{\mathrm{BS}}|$ against $\log n$. The expected behaviour is a cloud of points lying around a line of slope $-\frac{1}{2}$. To stabilise the plot against the variability of a single run, average $|\hat{C}_n - C_{\mathrm{BS}}|^2$ over independent seeds at each n and plot $\frac{1}{2}\log\big(\overline{|\hat{C}_n - C_{\mathrm{BS}}|^2}\big)$ against $\log n$, fitting a least-squares line to the resulting points. The fitted slope should be close to $-\frac{1}{2}$.

Test 3: agreement of sample variance with closed form. For the same parameter configurations, compare the sample variance S_n^2 produced by the simulation with the closed-form value delivered by Proposition 5.1. The relative error should decrease at the rate predicted by the variance of the sample variance estimator, namely $O(n^{-1/2})$. A persistent discrepancy at a level inconsistent with this rate indicates a bug in either the sampling scheme or the closed-form computation; the test discriminates between these two failure modes by comparison with a third independent computation, performed at very large n.

Test 4: monotonicities and asymptotic limits. The Black–Scholes price is monotone non-decreasing in S_0, in σ, and in T (for fixed strike and rate); it is monotone non-increasing in K. The Monte Carlo estimator must respect these monotonicities up to statistical noise, which can be enforced sharply by the use of common random numbers across parameter values (the same seed produces the same sequence of Z_i in both runs, and the inequality holds path by path). The asymptotic limit $\lim_{S_0\to\infty} C/S_0 = 1$ should also be approximated, as should $\lim_{S_0\to 0} C = 0$.

7 Algorithmic specification

The algorithm is summarised below. The pseudocode is parametric in the random source (a generator object accepting a seed) to enforce the reproducibility discipline of Block 0, Section 2.5.

Algorithm 1 Vanilla Monte Carlo for the European call (exact GBM)

Require: Spot S_0, strike K, maturity T, rate r, volatility σ, sample size n, confidence level $1-\beta$, random generator `rng` initialised with a recorded seed.

Ensure: Estimate $\hat{C}_n$, half-width h_n, sample variance S_n^2.

1: $\mu \leftarrow (r - \frac{1}{2}\sigma^2)\,T$ ▷ drift of $\log S_T$
2: $s \leftarrow \sigma\sqrt{T}$ ▷ stdev of $\log S_T$
3: $D \leftarrow e^{-rT}$ ▷ discount factor
4: Draw $Z_1, \ldots, Z_n \sim N(0,1)$ independently using `rng`.
5: **for** $i = 1$ to n **do**
6: $\quad S^{(i)} \leftarrow S_0 \exp(\mu + s\,Z_i)$
7: $\quad Y_i \leftarrow D \cdot \max(S^{(i)} - K,\, 0)$
8: **end for**
9: $\hat{C}_n \leftarrow \frac{1}{n}\sum_i Y_i$
10: $S_n^2 \leftarrow \frac{1}{n-1}\sum_i (Y_i - \hat{C}_n)^2$
11: $h_n \leftarrow z_{\beta/2}\sqrt{S_n^2/n}$
12: **return** $(\hat{C}_n,\, h_n,\, S_n^2)$

The implementation will compute steps 5–8 in vectorised form on the entire vector $(Z_1, \ldots, Z_n)$ at once, and the pseudocode is written elementwise only for clarity. In neither Python nor C++ should this loop appear as an explicit Python or C++ `for`: in Python the entire computation is a sequence of NumPy operations, and in C++ it is a single contiguous loop with no allocation inside.

8 Looking ahead

Three extensions of this algorithm structure the remainder of Phase 2.

Block 1.2: discretisation. The exactness of the sampling scheme (7) relies on the closed-form solution (6) of the SDE, which is unavailable for almost every model of practical interest beyond geometric Brownian motion. In the next algorithm of the phase, the closed form is replaced by a time-discretised approximation (Euler–Maruyama, then Milstein), and systematic error of order Δt or $\sqrt{\Delta t}$ enters alongside the statistical error analysed here. The bias–variance tradeoff of Section 5 of the Block 0 writeup becomes operational at that point, and the optimal allocation of the computational budget between sample size and time-step refinement is the central new question.

Block 2: variance reduction. Each technique introduced in Block 2—antithetic variates, control variates, stratified sampling, importance sampling—is implemented as a wrapper around the same estimator structure developed here, and is benchmarked against the present algorithm. The denominator of the variance reduction factor is Proposition 5.1; the numerator is the sample variance of the new estimator. The structural cleanliness of the present algorithm—a single sampling step, a single payoff evaluation, no per-replication state—makes it the natural reference implementation in this comparison.

Block 4: greeks. The pathwise and likelihood-ratio estimators of price sensitivities reuse the sample paths $\{S_T^{(i)}\}_{i=1}^n$ generated by the present algorithm, applying to them a different functional in place of the discounted payoff. The common-random-numbers technique used in conjunction with finite-difference greeks requires that two parameter-perturbed estimators consume the same sequence of Z_i, a requirement that the seed-controlled sampling discipline of this algorithm makes immediate.

9 Bibliographic notes

The exact-simulation Monte Carlo estimator for the European call under geometric Brownian motion is treated, as a textbook starting example, in Glasserman [1, Section 1.1.2 and Section 3.2]. The closed-form variance computation of Proposition 5.1 is elementary and appears implicitly in many sources; an explicit derivation in the same spirit as the present one can be found in Jäckel [2, Chapter 3]. The variance reduction factor as the figure of merit for Monte Carlo techniques is discussed in Glasserman [1, Section 4.1].

References

[1] P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer, 2004.

[2] P. Jäckel, *Monte Carlo Methods in Finance*, Wiley, 2002.

[3] A. Etheridge, *A Course in Financial Calculus*, Cambridge University Press, 2002.